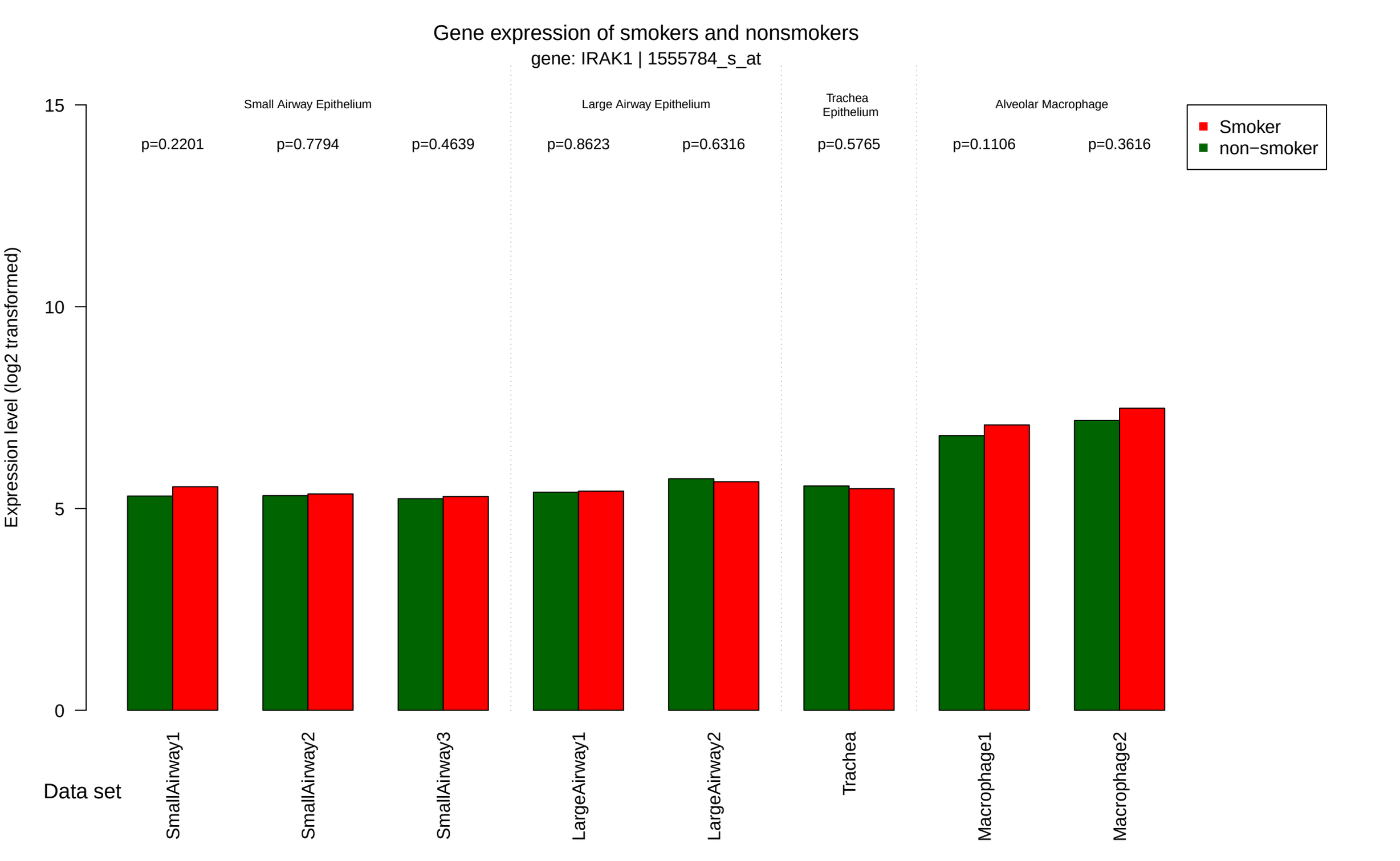
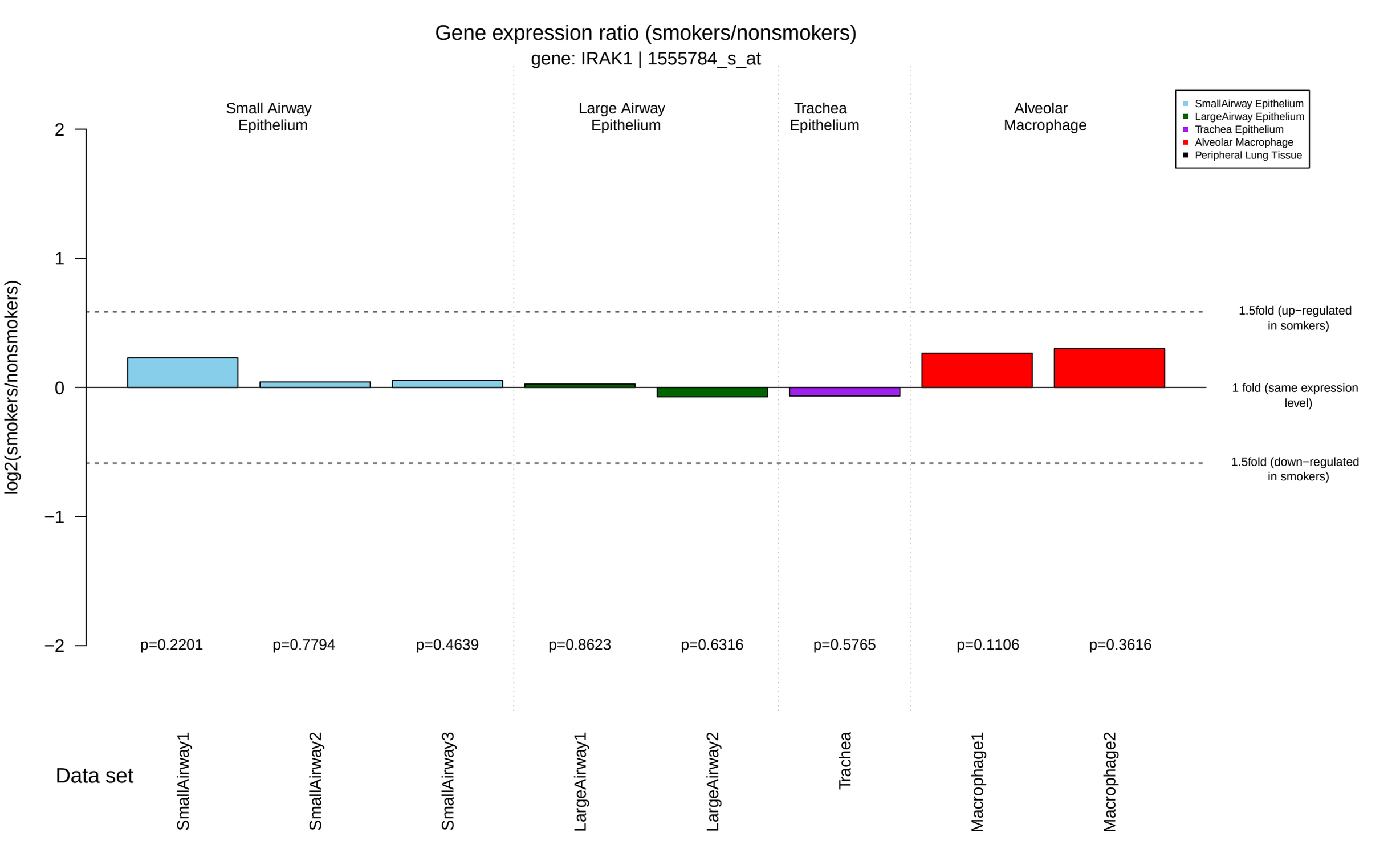
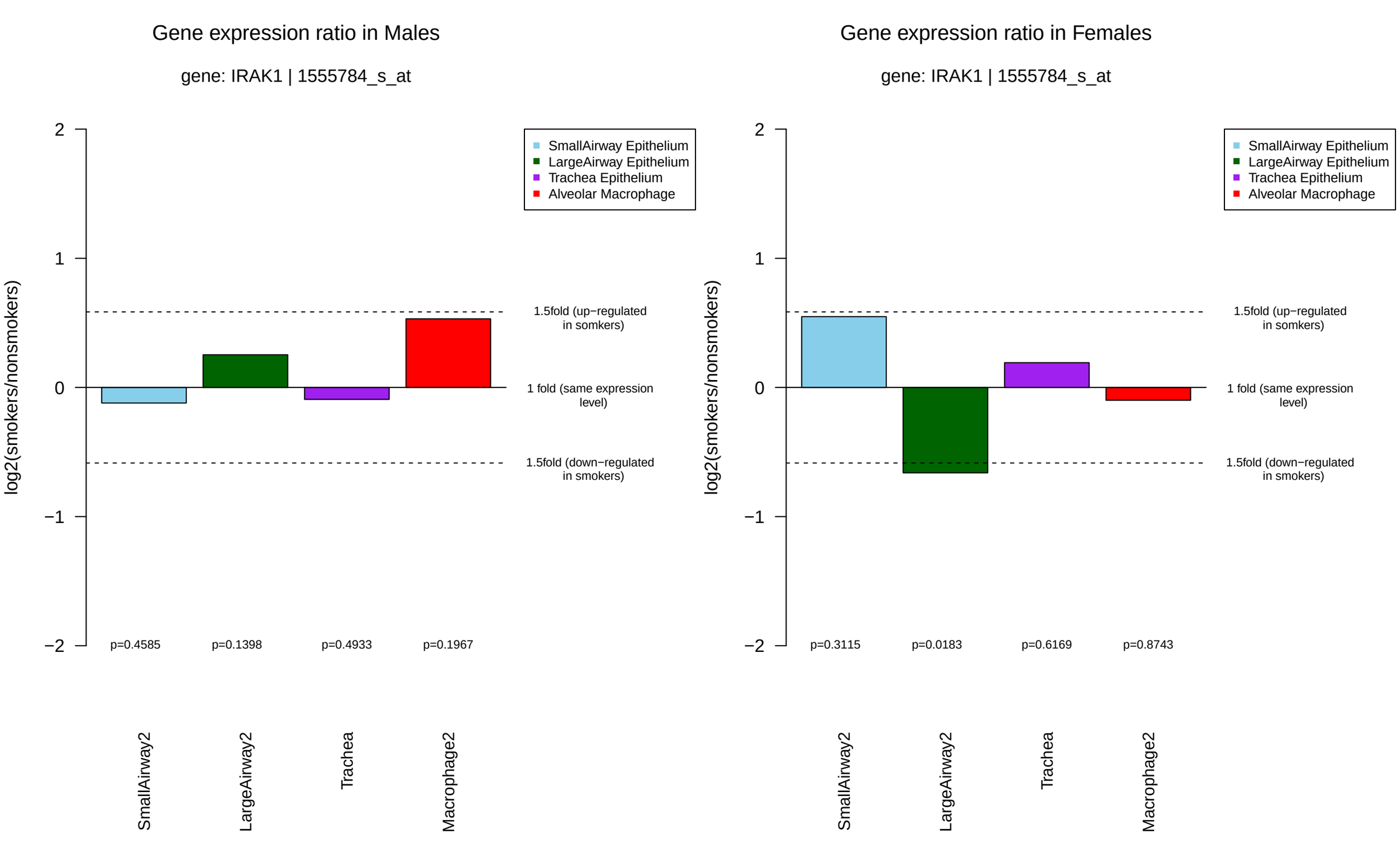
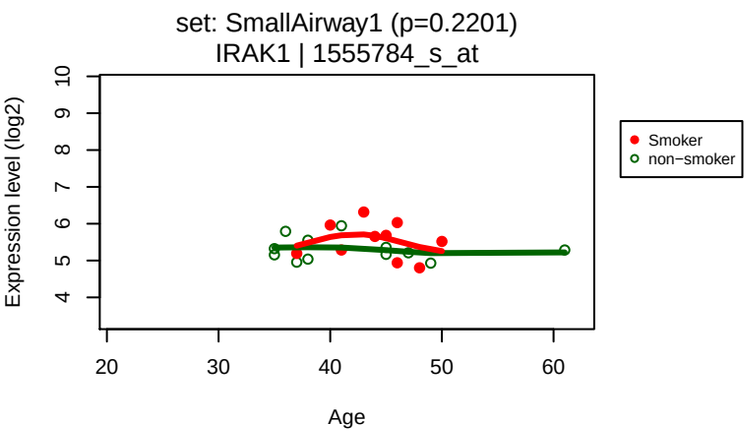
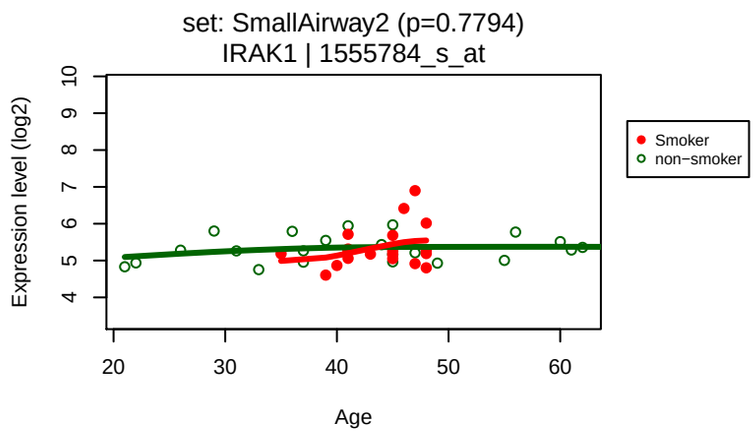
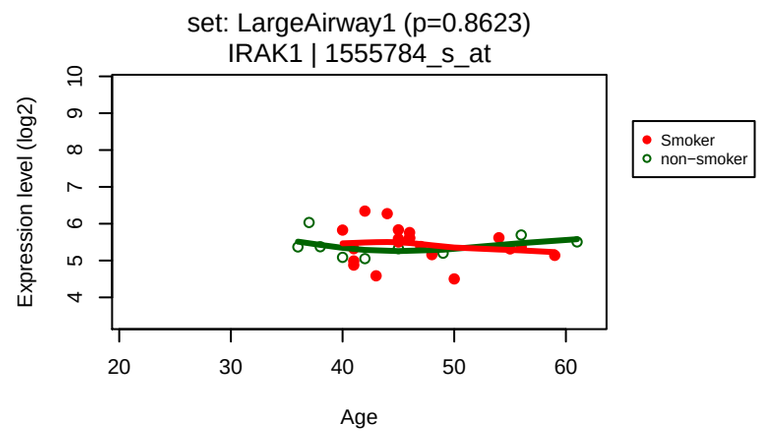

Gene expression of smokers and nonsmokers
gene: IRAK1 | 1555784_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2201
p=0.7794
p=0.4639
p=0.8623
p=0.6316
p=0.5765
p=0.1106
p=0.3616
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

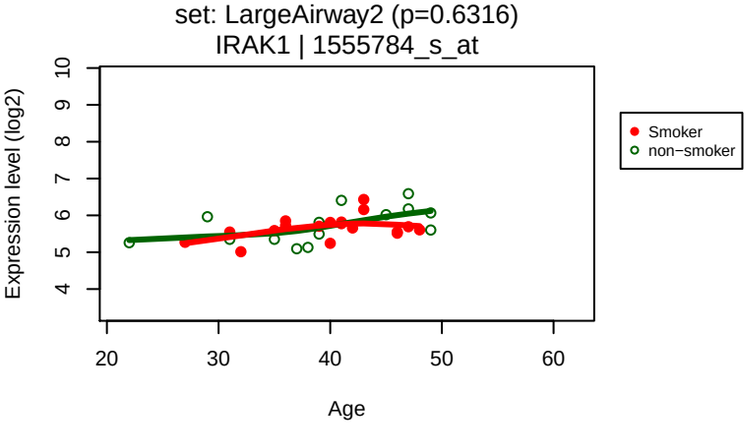

Gene expression ratio (smokers/nonsmokers)
gene: IRAK1 | 1555784_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2201
p=0.7794
p=0.4639
p=0.8623
p=0.6316
p=0.5765
p=0.1106
p=0.3616
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

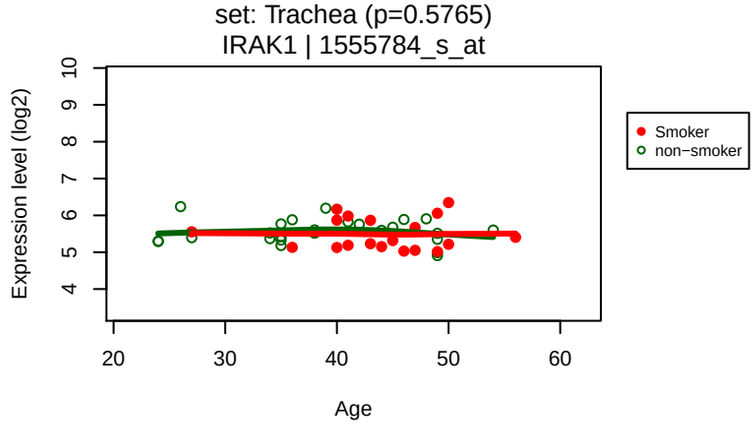

Gene expression ratio in Males
gene: IRAK1 | 1555784_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4585
p=0.1398
p=0.4933
p=0.1967
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRAK1 | 1555784_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3115
p=0.0183
p=0.6169
p=0.8743
SmallAirway2
LargeAirway2
Trachea
Macrophage2

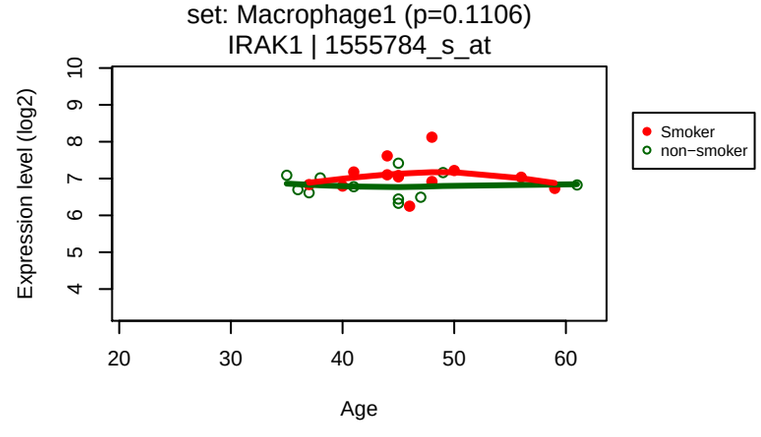

set: SmallAirway1 (p=0.2201)
IRAK1 | 1555784_s_at
Expression level (log2)
Age
Smoker
non−smoker

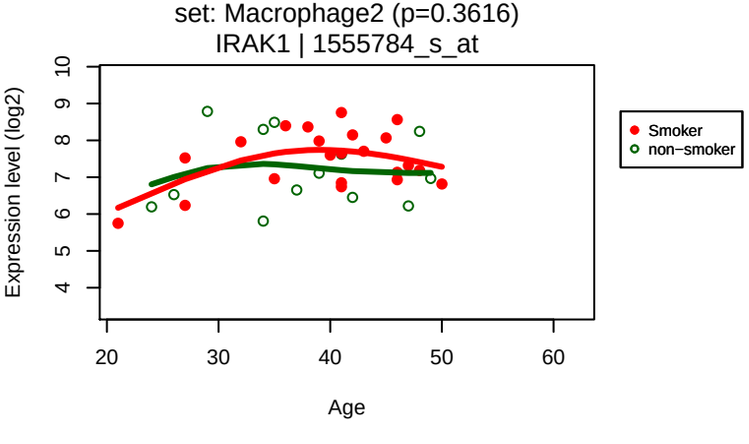

set: SmallAirway2 (p=0.7794)
IRAK1 | 1555784_s_at
Expression level (log2)
Age
Smoker
non−smoker

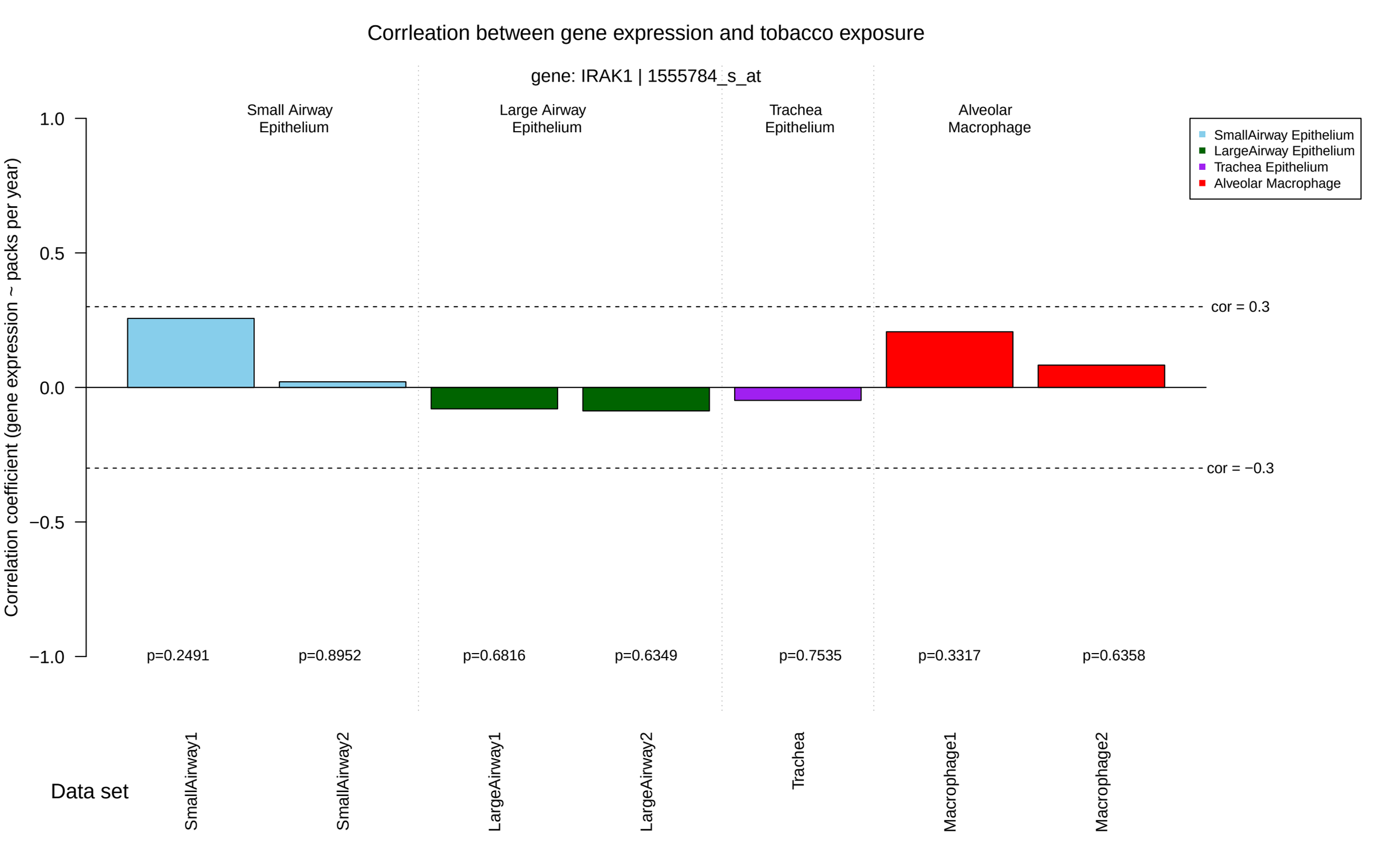

set: LargeAirway1 (p=0.8623)
IRAK1 | 1555784_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6316)
IRAK1 | 1555784_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5765)
IRAK1 | 1555784_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1106)
IRAK1 | 1555784_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3616)
IRAK1 | 1555784_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRAK1 | 1555784_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2491
p=0.8952
p=0.6816
p=0.6349
p=0.7535
p=0.3317
p=0.6358
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2